# An Audited Evaluation of Event-Aware Retrieval for 24-Hour PM2.5 Forecasting

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***Abstract*—Retrieval augmentation offers a practical way to supply forecasting models with historical analogues, but its value can be overstated when the memory is sparse, the comparison baseline is weak, or contextual channels are not tested independently. This study audits Event-TimeRAF, a retrieval framework comprising a Source-Audited Context Builder, a Leakage-Safe Event-Context Retriever, and a Drift-Aware Forecast and Evidence Head. EPA AirData PM2.5, NOAA Global Hourly weather, calendar variables, and hash-verified NOAA Storm Events records are aligned for Los Angeles County from 2019 through 2024. Evaluation uses a 168-hour lookback, a 24-hour horizon, 7,199 test origins, four knowledge-base strides, event-weight controls, retrieval placebos, and overlap-aware paired inference. The strongest model is the weather-calendar XGBoost baseline, with MSE 26.185, MAE 3.125, RMSE 5.117, and $R^2$ 0.379. The full Event-TimeRAF model obtains MSE 26.734, and its difference from the context baseline is unresolved. Frozen Chronos-Bolt obtains MSE 28.941; retrieval fusion obtains 28.733, but a climatology fusion control obtains 27.450 and significantly outperforms the retrieval fusion. A denser six-hour memory improves retrieval-only MSE to 34.616, yet the corresponding feature model remains behind the context baseline. Increasing event weight changes 91.4% of test top-$k$ sets but worsens retrieval error, indicating that the tested event representation does not add forecast value. Because Storm Events lacks machine-readable publication timestamps, event-conditioned findings remain retrospective sensitivity results.**

***Index Terms*—Air quality forecasting, PM2.5, retrieval-augmented forecasting, time-series foundation models, event context, distribution-shift diagnostics, forecast traceability.**

## I. Introduction

Fine particulate matter (PM2.5) is a major urban air-quality risk because it is associated with respiratory and cardiovascular health effects [1]. Accurate short-term PM2.5 forecasts can support public-health alerts, exposure reduction, transport planning, and environmental operations. In Los Angeles County, however, PM2.5 dynamics are difficult to model with pollutant history alone. Concentrations can change with wind speed, boundary-layer behavior, rainfall, seasonal activity, holidays, wildfire smoke, and unusual storm or high-wind events. These factors make the forecasting problem both temporal and contextual.

Recent time-series foundation models (TSFMs) aim to improve forecasting generalization by pre-training on large time-series corpora and transferring the learned forecasting ability to new domains. Representative models include TimesFM, Moirai, MOMENT, Chronos, and Tiny Time Mixers [2]–[6]. In parallel, retrieval-augmented generation has shown that nonparametric memory can help foundation models access relevant external knowledge at inference time [7], [8]. TimeRAF combines these two ideas for zero-shot time-series forecasting by retrieving relevant historical sequences from a curated knowledge base and integrating them into a frozen TSFM through Channel Prompting [9].

Air-quality forecasting raises a domain-specific retrieval question that is not fully addressed by numerical time-series retrieval alone. A historical PM2.5 window may look similar to the current window but occur under different weather or event conditions. Conversely, a window with moderate numerical similarity may be more informative if it shares high-wind, rainfall, holiday, smoke, or stagnation context. This motivates a retrieval framework that treats source records, weather, calendar information, and distribution-shift indicators as part of the forecasting evidence rather than as unrelated metadata.

This study evaluates Event-TimeRAF, an event-aware retrieval-augmented framework for PM2.5 forecasting under distribution shift. The method adapts the retrieval principle of TimeRAF to an official-source environmental setting without claiming to reproduce its learned retriever or Channel Prompting mechanism. Event-TimeRAF is organized into three named components: a Source-Audited Context Builder (SACB), a Leakage-Safe Event-Context Retriever (LSER), and a Drift-Aware Forecast and Evidence Head (DFEH). Together, these components address three questions: how official environmental records can be converted into a retrieval-compatible context; how historical candidates can be selected without temporal leakage; and when retrieval contributes beyond a strong supervised weather-calendar baseline.

The main contributions are as follows:

- *Source-Audited Context Builder:* a reproducible mechanism that aligns EPA PM2.5, NOAA weather, calendar, and hash-verified NOAA Storm Events records while preserving source identity, coverage checks, and the declared event-availability assumption.
- *Leakage-Safe Event-Context Retriever:* a historical analogue mechanism that enforces a query–candidate embargo and is tested across memory density, neighbor count, event weight, and event-stratified controls.
- *Drift-Aware Forecast and Evidence Head:* a controlled evaluation layer that compares direct XGBoost, frozen Chronos-Bolt, retrieval fusion, and non-retrieval fusion placebos while retaining machine-readable shift and provenance evidence.

The evaluation is deliberately conservative. It restricts claims to a single manifest-backed run, uses 2,000-resample paired block intervals and overlap-aware Diebold–Mariano tests, and does not interpret event association as causation. The resulting negative findings distinguish a functioning, au-

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ditable retrieval pipeline from evidence that retrieval improves forecasting.

## II. Literature Review and Related Work

### A. Statistical, Machine-Learning, and Deep Forecasters

Forecasting methods differ in how they trade inductive bias, data demand, and computational cost. Autoregressive and seasonal models remain informative baselines because their failure modes are easy to inspect. Tree ensembles such as XGBoost and LightGBM extend this setting to nonlinear lag, rolling, weather, and calendar interactions without requiring a sequence encoder [10], [11]. Recurrent LSTM and GRU models instead learn stateful temporal representations [12], [13]. Large benchmark archives, including the Monash Forecasting Repository, further support reproducible comparison across heterogeneous time-series domains [14]. Transformer families emphasize longer dependencies: Informer reduces attention cost, Autoformer and FEDformer use decomposition, Crossformer and iTransformer model cross-variable structure, and TimesNet reorganizes temporal variation into two-dimensional patterns [15]–[20]. These gains do not imply that greater architectural complexity is always necessary. DLinear and PatchTST show that simple decomposition or patch-based inductive biases can remain highly competitive [21], [22], whereas TimeMixer uses explicit multiscale decomposition and mixing [23]. Event-TimeRAF therefore retains seasonal and boosted tree baselines instead of treating a deep backbone as the default winner.

### B. Time-Series Foundation Models

Time-series foundation models (TSFMs) seek transfer across datasets rather than training a separate representation from scratch for each series. TimesFM uses a decoder-only forecasting design; Moirai trains a universal forecasting Transformer; MOMENT supports several time-series tasks; Chronos represents values as tokens; and Tiny Time Mixers emphasizes compact zero- and few-shot transfer [2]–[6]. Timer, Timer-XL, UniTime, and Time-LLM explore generative pretraining, long-context scaling, cross-domain unification, or language-model reprogramming [24]–[27]. Frozen pretrained models have also been evaluated as general computation engines for time-series analysis [28]. Collectively, these studies motivate zero-shot evaluation, but they also show that transfer performance depends on input representation and domain shift. The present study therefore compares frozen Chronos-Bolt against locally supervised models on identical forecast origins rather than assuming foundation-model superiority.

### C. Retrieval-Augmented Forecasting

Retrieval augmentation separates parametric knowledge from an external memory. In natural-language processing, RAG and Dense Passage Retrieval established the retrieve-then-condition pattern [7], [8]. In time series, retrieval has been studied through diffusion forecasting and assisted generation workflows [29], [30]. TimeRAF provides the closest methodological reference: it constructs a time-series knowledge base, trains an MLP dual-encoder retriever, selects top-$k$ candidates, and integrates candidate embeddings into a frozen TSFM through Channel Prompting [9]. Its ablations separate retrieval quality, integration, candidate count, knowledge-base composition, and backbone choice.

Event-TimeRAF retains the external-memory principle but addresses a different experimental question. It uses transparent random, cosine, and event-context similarity instead of a learned dual encoder, and it uses feature- or forecast-level fusion instead of internal Channel Prompting. This distinction prevents the current implementation from being described as a reproduction of TimeRAF; it is an audited environmental adaptation designed to test whether contextual historical analogues help a local PM2.5 task.

### D. Air-Quality Forecasting and Official Context

PM2.5 forecasting commonly combines pollutant history with temperature, humidity, wind, precipitation, and pressure. Recent reviews describe a shift from single-station recurrent models toward attention-based, hybrid, and spatiotemporal systems [31], [32]. AirFormer, for example, separates deterministic spatiotemporal modeling from stochastic uncertainty estimation at nationwide scale [33]. Such multi-station designs address spatial dependence, whereas the present study examines a county-level aggregate and the provenance of contextual evidence. EPA AirData supplies regulated PM2.5 observations, NOAA Integrated Surface Database supplies hourly meteorology, and NOAA Storm Events supplies structured hazard records [34]–[36]. NOAA Hazard Mapping System products could add smoke evidence but were not included in the reported run [37].

### E. Distribution Shift and Forecast Explanations

Temporal nonstationarity can alter means, variances, and relationships between inputs and future values. Adaptive normalization methods address this problem by estimating statistics at local temporal scales [38]; Event-TimeRAF instead uses interpretable shift indicators to audit regimes without claiming that shift has been eliminated. Explanation methods also differ in scope. SHAP attributes predictions to model inputs [39], while information-theoretic saliency supports counterfactual inspection of probabilistic forecasts [40]. The implemented explanation layer is narrower: it records signed XGBoost effects, retrieved cases, event identifiers, drift components, and a diagnostic uncertainty scale. These records support traceability, but they do not establish causal effects of weather or events.

## III. Critical Gaps and Limitations of Previous Studies

Prior work leaves three connected gaps for event-sensitive urban air-quality forecasting. First, temporal forecasters and numerical retrieval systems do not necessarily preserve the provenance, coverage, and availability assumptions of local contextual records. A numerically similar PM2.5 window can occur under a different weather or event regime, and a retrospective event label can be mistaken for information available at forecast time. The SACB addresses this gap by joining official records under explicit source, coverage, and availability audits.

Second, historical retrieval can introduce look-ahead bias when a candidate input or target overlaps the query context. Similarity alone does not prevent a near-duplicate future from entering the retrieved evidence. The LSER addresses this gap through a strict temporal embargo and training-history restriction, followed by separately auditable PM2.5, weather, calendar, and event similarity terms.

Third, retrieval gains are often reported without testing whether a strong local context model already captures the useful signal, whether the gain persists under distribution shift, or whether an explanation can be traced to stored evidence. The DFEH addresses this gap by evaluating direct XGBoost and frozen Chronos-Bolt paths on common origins, exposing shift indicators, and retaining machine-readable feature, retrieval, event, and uncertainty evidence. This component supports comparison and auditability; it does not convert contextual association into causal explanation.

These mappings delimit the novelty of the study. Event-TimeRAF does not contain a learned dual-encoder retriever or internal Channel Prompting, and the present single-county experiment does not establish geographic generalization.

## IV. Core Contributions

The proposed framework contributes three mechanisms corresponding one-to-one with the gaps above:

- *Source-Audited Context Builder (SACB):* constructs an aligned 85-dimensional context from official PM2.5, meteorological, calendar, and event records while retaining source and availability metadata.
- *Leakage-Safe Event-Context Retriever (LSER):* retrieves training-history analogues only after enforcing a candidate-target embargo, then exposes the contribution of each similarity channel and the retrieved trajectory evidence.
- *Drift-Aware Forecast and Evidence Head (DFEH):* places direct horizon models, frozen-TSFM forecast fusion, drift-state analysis, and traceable explanation records within one chronological evaluation contract.

The empirical contribution is intentionally qualified: weather-calendar context produces the clearest reduction in error, denser retrieval improves only the retrieval-only path, and the tested event channel does not improve forecast accuracy.

## V. Problem Formulation

Let $x_t$ denote the PM2.5 measurement at time $t$. For a forecast time $t$, the historical air-quality lookback window is

$$X_t = [x_{t-L+1}, x_{t-L+2}, \ldots, x_t], \tag{1}$$

where $L$ is the lookback length. Let $W_t$ denote weather covariates aligned with the lookback window, $C_t$ denote calendar features, and $E_t$ denote event context available under the recorded source assumption at the forecast origin. The target associated with the input in (1) is

$$Y_{t+1:t+H} = [x_{t+1}, x_{t+2}, \ldots, x_{t+H}], \tag{2}$$

where $H$ is the forecasting horizon. Equation (2) therefore defines the 24-step vector evaluated at every valid origin.

The forecasting goal is to learn a function

$$\hat{Y}_{t+1:t+H} = F_\theta(X_t, W_t, C_t, E_t, R_t^{ts}, R_t^{ev}, D_t), \tag{3}$$

where $R_t^{ts}$ is retrieved time-series knowledge, $R_t^{ev}$ is retrieved event knowledge, and $D_t$ contains concept-drift indicators. The implemented target setting is $L = 168$ hours and $H = 24$ hours. Equation (3) states the full information interface; the implemented modules that construct these arguments are specified in the next section.

The evaluation uses a time-based split:

$$\mathcal{T}_{train} < \mathcal{T}_{val} < \mathcal{T}_{test}, \tag{4}$$

where the earliest 70% of observations are used for training, the next 15% for validation, and the latest 15% for testing. The ordering in (4) is retained for feature fitting, retrieval eligibility, model selection, and final evaluation.

Table I summarizes the essential notation.

TABLE I
Main Symbols Used in the Proposed Formulation

| Symbol | Meaning |
|---|---|
| $L$ | Lookback window length |
| $H$ | Forecast horizon |
| $X_t$ | Historical PM2.5 sequence at time $t$ |
| $W_t$ | Weather covariates |
| $C_t$ | Calendar/context features |
| $E_t$ | Event context available at the forecast origin |
| $R_t^{ts}$ | Retrieved time-series knowledge |
| $R_t^{ev}$ | Retrieved event knowledge |
| $D_t$ | Drift indicators |
| $q_t$ | SACB context vector in $\mathbb{R}^{85}$ |
| $G_t$ | LSER top-$k$ retrieved set |
| $\rho(G_t)$ | Retrieval summary in $\mathbb{R}^{51}$ |
| $\mathcal{O}_t$ | Machine-readable DFEH evidence record |
| $\hat{Y}_{t+1:t+H}$ | Predicted future PM2.5 values |

## VI. Methodology

Event-TimeRAF comprises three top-level components. The Source-Audited Context Builder (SACB) converts heterogeneous official records into a chronological hourly context while retaining provenance and availability metadata. The Leakage-Safe Event-Context Retriever (LSER) constructs an embargoed historical memory and selects analogues according to PM2.5, weather, calendar, and event similarity. The Drift-Aware Forecast and Evidence Head (DFEH) exposes feature-level and forecast-level prediction paths, shift indicators, and a machine-readable explanation record. These components are deterministic or supervised modules over one-dimensional time-series and tabular inputs; no image operator, learned attention block, or internal TimeRAF Channel Prompting is implied.

### A. Framework Overview and Forward Pass

Figure 1 summarizes the complete dataflow. EPA PM2.5, NOAA weather, calendar, and NOAA Storm Events records first enter the SACB, which audits, aligns, and transforms them into a query context and input–target windows. The LSER filters the training-history knowledge base, ranks eligible candidates, and returns aligned candidate futures and retrieval statistics. The DFEH then supplies those statistics to 24 direct XGBoost regressors, evaluates a separate frozen Chronos-Bolt fusion path, computes drift evidence, and writes the forecast and its supporting record.

For each forecast origin $t$, Event-TimeRAF maps the aligned context into a forecast through

$$\hat{Y}_{t+1:t+H} = F_\theta\big(\phi(X_t, W_t, C_t, E_t), G_t, D_t\big), \tag{5}$$

where $\phi(\cdot)$ denotes the tabular feature construction, $G_t$ denotes retrieved historical evidence, and $D_t$ denotes drift evidence. The key methodological constraint is that every element of $G_t$ and $D_t$ must be available no later than the forecast origin. Equation (5) is the compact functional form that the tensor-level forward pass below expands.

For a batch of $B$ origins, the implemented forward pass can be expanded in tensor notation as

$$\begin{aligned}
Q &= \mathrm{SACB}(X, W, C, E) && \in \mathbb{R}^{B\times 85},\\
G &= \mathrm{LSER}(Q, \mathcal{K}) && \in \mathbb{R}^{B\times 8\times 24},\\
U_h &= [Q \,\|\, \rho(G) \,\|\, d \,\|\, C_h^+] && \in \mathbb{R}^{B\times 151},\\
\hat{Y}^X &= [f_1(U_1), \ldots, f_{24}(U_{24})] && \in \mathbb{R}^{B\times 24},\\
(\hat{Y}, \mathcal{O}) &= \mathrm{DFEH}(\hat{Y}^X, \hat{Y}^C, \hat{Y}^R, d, G) && \in \mathbb{R}^{B\times 24} \times \mathcal{E},
\end{aligned} \tag{6}$$

where $Q$ is the 85-dimensional origin context, $\rho(G) \in \mathbb{R}^{B\times 51}$ contains the retrieved trajectory, spread, two similarity summaries, and candidate count, $d \in \mathbb{R}^{B\times 6}$ contains five drift components and their mean score, and $C_h^+ \in \mathbb{R}^{B\times 9}$ is the known calendar vector for horizon $h$. The symbol $\mathcal{E}$ denotes the schema of the evidence record $\mathcal{O}$. Equation (6) describes both reported forecasting paths without presenting the XGBoost path as part of the frozen foundation model.

### B. Source-Audited Context Builder

The SACB is motivated by the need to preserve data provenance and temporal availability before model fitting. As shown in Figure 2, source-specific checks precede feature construction. EPA monitor records are aggregated to an hourly county median, NOAA weather is aligned from the selected surface station, event archives are hash-verified against a manifest, and calendar values are derived from the local timestamp. The resulting context is then windowed and split chronologically; target values are never interpolated.

*1) Dataset Construction:* Hourly PM2.5 observations are taken from US EPA AirData/AQS parameter 88101 for Los Angeles County [34]. The data audit selects one monitoring site using coverage and continuity criteria when a single site satisfies the coverage criterion. Because no individual monitor met this criterion, the target is the hourly county median across official Los Angeles County AQS monitors. Timestamps are converted to `America/Los_Angeles` with explicit daylight-saving handling. Short input gaps may be filled only from past-available observations; targets are never interpolated. Weather covariates come from the nearest suitable NOAA Integrated Surface Database station [35]. Calendar features include cyclic hour, weekday, and month terms, weekend status, US federal holidays, and California holidays.

At each origin, the feature groups are concatenated as

$$q_t = [p_t \,\|\, w_t \,\|\, c_t \,\|\, e_t] \in \mathbb{R}^{23+14+9+39} = \mathbb{R}^{85}, \tag{7}$$

where $p_t$, $w_t$, $c_t$, and $e_t$ are respectively the PM2.5, weather, calendar, and event feature vectors. The PM2.5 group contains fixed lags, rolling moments and extrema, differences, the current value, and a fill flag. The weather group contains six measured variables, a precipitation-missing flag, derived condition flags, a stagnation proxy, and 24-hour rolling means. The event group contains one-hour and active counts, 24/72/168-hour counts, category-specific 72-hour counts, and an event-burst ratio. Thus, Equation (7) fixes both the feature ordering and the 85-dimensional SACB output used by downstream modules.

All normalization statistics are fitted on the training split only. Query and candidate windows use identical normalization, following the TimeRAF principle that retrieved and input sequences must share a preprocessing convention. The reported run contains 48,332 valid windows with $X \in \mathbb{R}^{48332\times 168}$ and $Y \in \mathbb{R}^{48332\times 24}$.

### C. Leakage-Safe Event-Context Retriever

The LSER separates temporal eligibility from relevance. Figure 3 first shows the embargo: a candidate is considered only if its complete target precedes the beginning of the query lookback. The eligible candidates are then compared along four independently stored similarity channels. The selected futures are aligned to the query scale and summarized for the forecasting head. This ordering prevents a high similarity score from overriding the temporal constraint.

*1) Time-Series Knowledge Base:* The time-series knowledge base contains historical PM2.5 windows. Each record stores a window identifier, start time, end time, input and future values, weather summary, calendar summary, normalized window vector, split, and normalization identifier. The primary configuration uses a 24-hour origin stride and contains 1,521 training-history candidates. Candidate windows may overlap one another; causality is instead enforced between every query and candidate by Equation (9). Sensitivity experiments use 192-, 24-, 6-, and 1-hour strides, producing 191, 1,521, 5,644, and 34,020 candidates, respectively.

Formally, the knowledge base is

$$\mathcal{K} = \{(u_i, X_i, Y_i, m_i, c_i, e_i)\}_{i=1}^{N}, \tag{8}$$

where $u_i$ is the candidate origin, $X_i$ and $Y_i$ are the candidate lookback and future windows, $m_i$ is a weather summary, $c_i$ is a calendar summary, and $e_i$ is the event-context summary available at $u_i$. For a query origin $t$, the temporally eligible set is

$$\mathcal{A}_t = \{i : u_i + H < t - L + 1,\; i \in \mathcal{T}_{train}\}, \tag{9}$$

for validation and test queries. The strict inequality in (9) ensures that a candidate's target period finishes before the

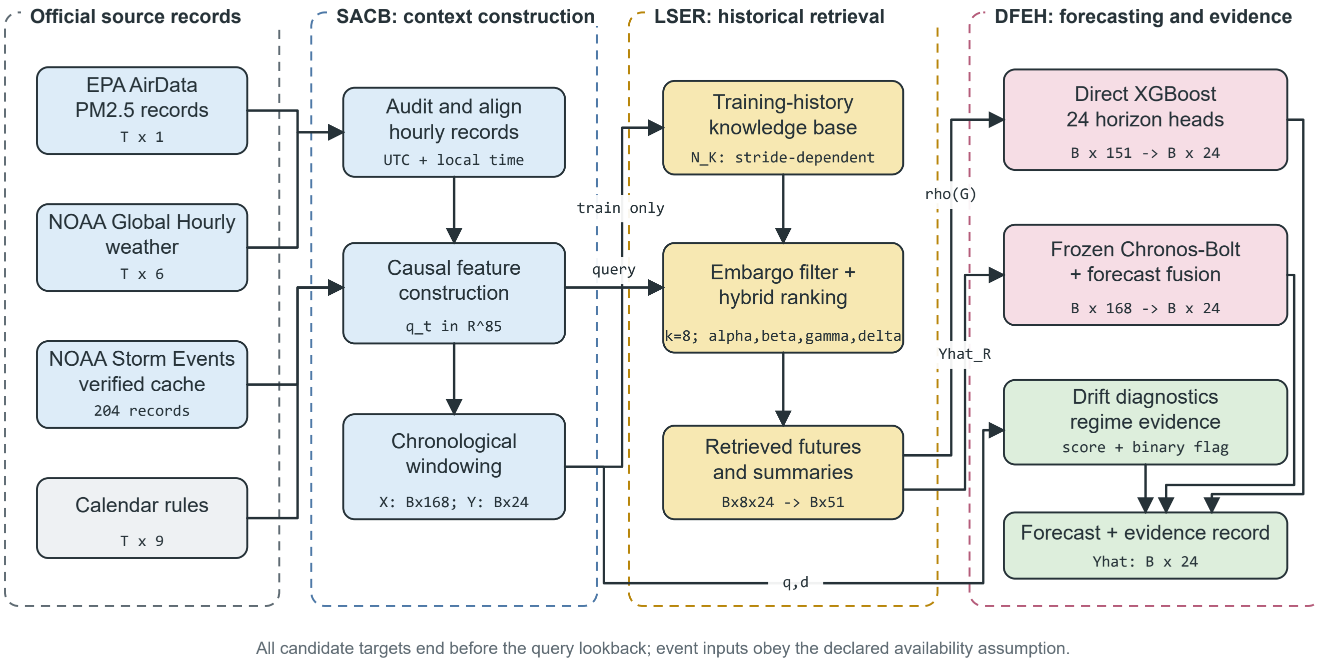


Fig. 1. Proposed Event-TimeRAF pipeline. Official records pass through the Source-Audited Context Builder (SACB), training-history candidates are selected by the Leakage-Safe Event-Context Retriever (LSER), and the Drift-Aware Forecast and Evidence Head (DFEH) produces 24-hour forecasts and traceable evidence. Tensor annotations show the reported configuration.

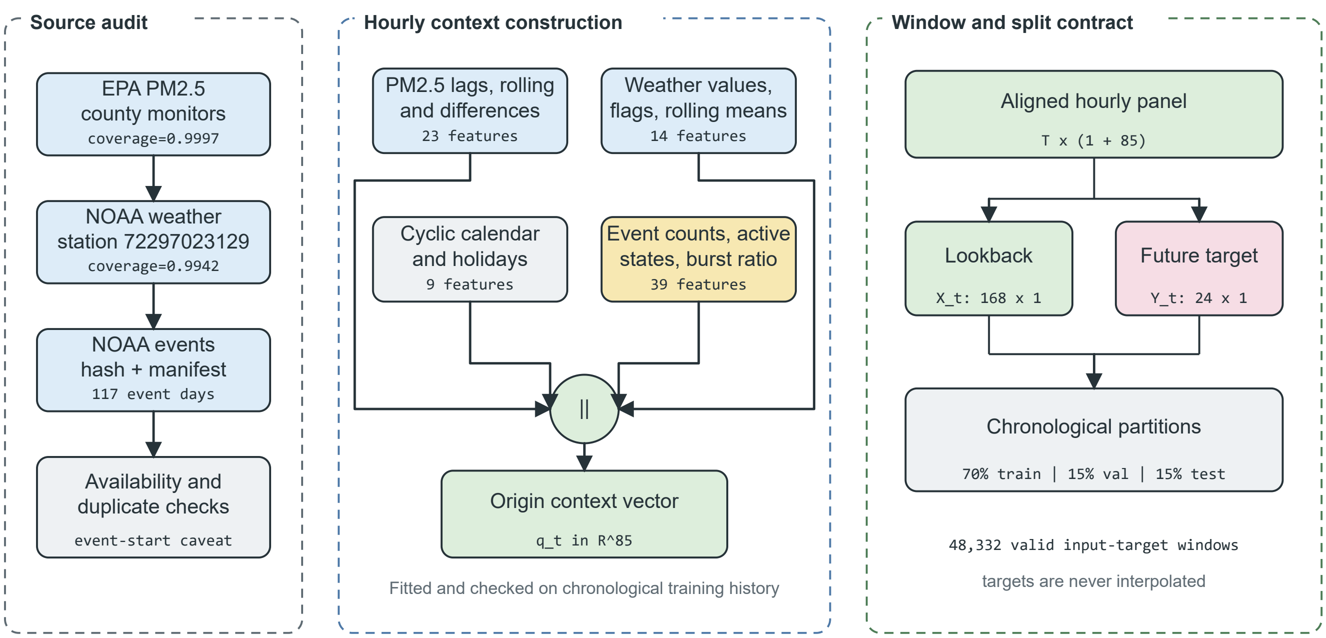


Fig. 2. Source-Audited Context Builder. Source checks feed four feature groups that form an 85-dimensional origin vector. The aligned panel is converted into 168-hour inputs and 24-hour targets before the chronological split.

query input period begins. The training-history condition in (9) is also enforced for final validation and test retrieval. Equation (8) defines the stored candidate schema, while Equation (9) defines the admissible subset at query time.

Given a query window $X_t$, the time-series retriever returns

$$R_t^{ts} = \{r_{t,1}^{ts}, r_{t,2}^{ts}, \ldots, r_{t,k}^{ts}\}, \tag{10}$$

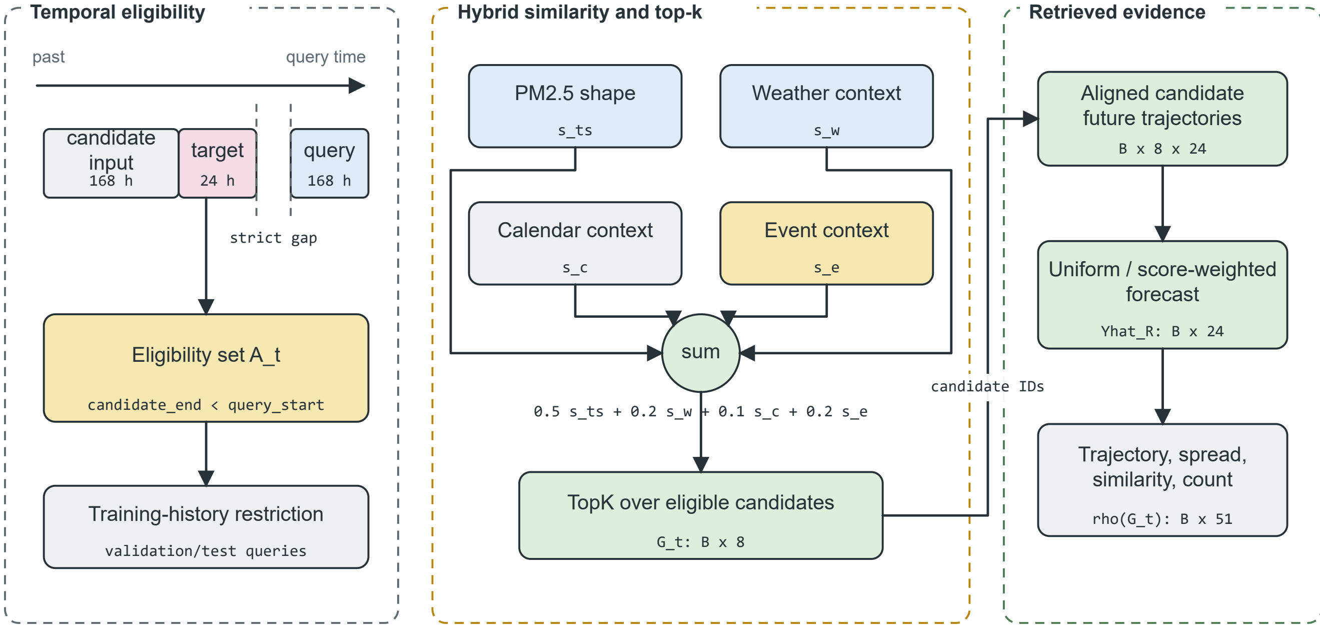


Fig. 3. Leakage-Safe Event-Context Retriever. A strict candidate-target embargo and training-history restriction are applied before hybrid ranking. The top eight candidates provide aligned future trajectories and a 51-dimensional retrieval summary.

where $k$ is the number of retrieved candidates. The default is $k = 8$, with sensitivity analysis for $k \in \{1, 4, 8, 16\}$. Equation (10) denotes the returned records before their future trajectories are summarized.

*2) Event Knowledge Base:* The event knowledge base contains source-preserving records relevant to Los Angeles County. NOAA Storm Events is the required structured source, whereas NOAA Hazard Mapping System fire and smoke records remain an unmeasured optional extension [36], [37]. Stored fields include event identifier, start and end, information-availability assumption, location, category, source, source URL, title, and summary. The reported records cover wildfire, high wind, heavy rain, flood, and other weather categories; feature slots for absent configured categories remain zero.

Model inputs represent events with hourly counts, rolling 24-, 72-, and 168-hour counts, category-specific 72-hour counts, active-event counts, and an event-burst ratio. Recent source identifiers are retained as explanation evidence. Target-horizon overlap is used only as a post-hoc subset label and never as a model input. NOAA Storm Events does not provide a machine-readable publication timestamp, so the implementation records event start as a retrospective availability assumption. Strict operational experiments require records with genuine publication or issue times.

*3) Hybrid Ranking and Future Alignment:* Both query and candidate PM2.5 windows are standardized by their own means and standard deviations and normalized to unit length. Weather and calendar groups are standardized using candidate statistics and mapped cosine similarity to $[0, 1]$. Event similarity is clipped to $[0, 1]$ with explicit zero-vector handling. The hybrid retrieval score combines time-series similarity, weather relevance, calendar relevance, and event relevance:

$$s(q, i) = \alpha s_{ts}(q, i) + \beta s_w(q, i) + \gamma s_c(q, i) + \delta s_e(q, i), \quad (11)$$

where $s_{ts}$ is time-series similarity, $s_w$ is weather similarity, $s_c$ is calendar similarity, and $s_e$ is event relevance. The implemented weights are $\alpha = 0.5$, $\beta = 0.2$, $\gamma = 0.1$, and $\delta = 0.2$. Equation (11) keeps each relevance channel explicit so that its contribution can be inspected without a learned latent scoring function.

The retrieved set is selected as

$$G_t = \mathrm{TopK}_{i \in \mathcal{A}_t} \; s(t, i). \quad (12)$$

Equation (12) is evaluated only after applying $\mathcal{A}_t$. Candidate futures are restored to the query scale as

$$\widetilde{Y}_{t,i} = \mu_t + \sigma_t \left( \frac{Y_i - \mu_i}{\max(\sigma_i, \epsilon)} \right), \qquad i \in G_t, \quad (13)$$

where $(\mu_t, \sigma_t)$ and $(\mu_i, \sigma_i)$ are the query and candidate lookback statistics, and $\epsilon = 10^{-6}$ prevents division by zero. Retrieval-only forecasts average $\widetilde{Y}_{t,i}$ uniformly or by non-negative normalized scores. The feature-level path retains the mean trajectory, horizon-wise spread, mean and maximum time-series similarity, and candidate count, yielding $\rho(G_t) \in \mathbb{R}^{51}$. The affine transformation in Equation (13) preserves the candidate trajectory shape while restoring the query's local scale.

### D. Drift-Aware Forecast and Evidence Head

The DFEH is the common output layer for two verified forecasting paths and the evidence output. As shown in Figure 4, the direct path concatenates context, retrieval, drift, and future-calendar features for 24 separately trained XGBoost regressors. A second path evaluates frozen Chronos-Bolt and combines its forecast with the retrieved forecast at a validation-selected weight. Drift scores and flags are retained as diagnostic inputs and evidence fields rather than being used to select between forecasters.

*1) Feature-Level Forecasting and Drift Diagnostics:* For horizon $h$, the direct model input and prediction are

$$u_{t,h} = [q_t \,\|\, \rho(G_t) \,\|\, d_t \,\|\, c_{t+h}], \qquad \hat{y}^X_{t+h} = f_h(u_{t,h}), \quad (14)$$

where $u_{t,h} \in \mathbb{R}^{151}$ for the full M09 model and each $f_h$ is an XGBoost regressor trained with squared-error objective. This design gives each forecast horizon a separate supervised function while retaining a common origin representation. Equation (14) also accounts for the verified 151 features: 85 context, 51 retrieval, six drift, and nine future-calendar values. The drift score and binary flag support regime analysis and the evidence record; they do not adaptively select a forecaster in the reported experiment.

*2) Frozen-TSFM Forecast Fusion:* The Chronos-Bolt retrieval variant uses forecast-level fusion rather than internal Channel Prompting. Let $\hat{Y}^C_t$ be the frozen Chronos forecast and $\hat{Y}^R_t$ be the retrieved historical forecast. The fused prediction is

$$\hat{Y}^{CR}_t(\omega) = \omega \hat{Y}^C_t + (1-\omega)\hat{Y}^R_t, \quad (15)$$

where $\omega \in \{0, 0.25, 0.5, 0.75, 1.0\}$ is selected on the validation split. Validation selected $\omega = 0.75$. Because fusion occurs after both forecasts are produced, this branch does not modify Chronos-Bolt embeddings or weights. Equation (15) is therefore an output-space interpolation, not an internal TSFM prompting operation.

*3) Concept-Drift Evidence:* The implementation represents distribution shift using transparent indicators rather than claiming perfect concept-drift identification. The indicators are recent mean shift, recent variance shift, retrieval similarity drop, weather-pattern shift, and event-burst ratio. The raw vector used by the code is

$$r_t = \Bigg[ |\mu_{24,t} - \mu_{168,t}|, \left| \log \frac{\sigma_{24,t} + \epsilon}{\sigma_{168,t} + \epsilon} \right|, 1 - \mathrm{clip}(\bar{s}_{ts,t}, 0, 1), \sqrt{\frac{1}{4} \sum_{\ell=1}^{4} \left( \frac{w_{t,\ell} - \widetilde{w}^{tr}_\ell}{s^{tr}_{w,\ell}} \right)^2 }, \ \max(b_{event,t}, 0) \Bigg], \quad (16)$$

where the weather term uses temperature, relative humidity, pressure, and wind speed standardized by training medians and standard deviations. Each component $r_{t,j}$ is then robustly scaled using the training median and median absolute deviation (MAD):

$$d_{t,j} = \frac{1}{6}\,\mathrm{clip}\left( \frac{r_{t,j} - \mathrm{med}_{tr}(r_j)}{\max(1.4826\,\mathrm{MAD}_{tr}(r_j), \epsilon)}, 0, 6 \right). \quad (17)$$

Finally, the continuous score and binary flag are

$$S_t = \frac{1}{5}\sum_{j=1}^{5} d_{t,j}, \qquad D_t = \mathbb{I}[S_t \geq Q_{0.90}(\{S_v : v \in \mathcal{T}_{val}\})]. \quad (18)$$

Equations (16)–(18) match the implemented detector: training data set the reference statistics, validation data set the 0.90-quantile threshold, and test data are transformed without refitting. The binary $D_t$ is retained for diagnostic regime analysis and explanation evidence.

*4) Evidence Record and Diagnostic Scale:* The explanation module uses signed local XGBoost contributions averaged across the 24 direct models, weather flags, the top three retrieved windows, recent event records, calendar context, forecast direction, and component-level drift evidence. The local values are XGBoost TreeSHAP contributions returned by `pred_contribs=True`; the bias column is removed before the values are averaged across horizons. Its diagnostic uncertainty scale is

$$u^{\mathrm{diag}}_t = \sqrt{\bar{\sigma}^2_{R,t} + \overline{\mathrm{RMSE}}^2_{val}}, \quad (19)$$

where $\bar{\sigma}_{R,t}$ is the mean horizon-wise spread of the retrieved trajectories and $\overline{\mathrm{RMSE}}_{val}$ is the mean of the 24 horizon-specific validation RMSE values. The result is a diagnostic scale, not a calibrated prediction interval. Each explanation is generated from stored fields rather than an unrestricted language model, and each event statement remains contextual rather than causal. Equation (19) is retained only as an evidence-field definition and is not used to construct confidence intervals.

### E. Relationship to TimeRAF

TimeRAF retrieves top-$k$ time-series windows using a learned MLP dual encoder and integrates candidate embeddings into a frozen TSFM through Channel Prompting [9]. Table II identifies which principles are retained and which mechanisms differ. The comparison is methodological rather than a performance ranking because the studies use different datasets, horizons, and protocols.

## VII. Experimental Setup

This section describes the final experiment. Model-selection choices were made using the training and validation partitions before test evaluation. Artifact consistency was verified using run identifier `20260810T103436161252Z`.

### A. Dataset Description and Audit

The study period is 2019–2024. PM2.5 observations were taken from US EPA AirData/AQS parameter 88101 for Los Angeles County [34]. No single monitoring site was used as the target series; the audited target is the hourly county median across official Los Angeles County AQS monitors. This fallback selected `06-037-COUNTY`, with 52,602 observed target hours and observed PM2.5 coverage of 0.9997. The PM2.5 unit was consistently recorded as micrograms per cubic meter.

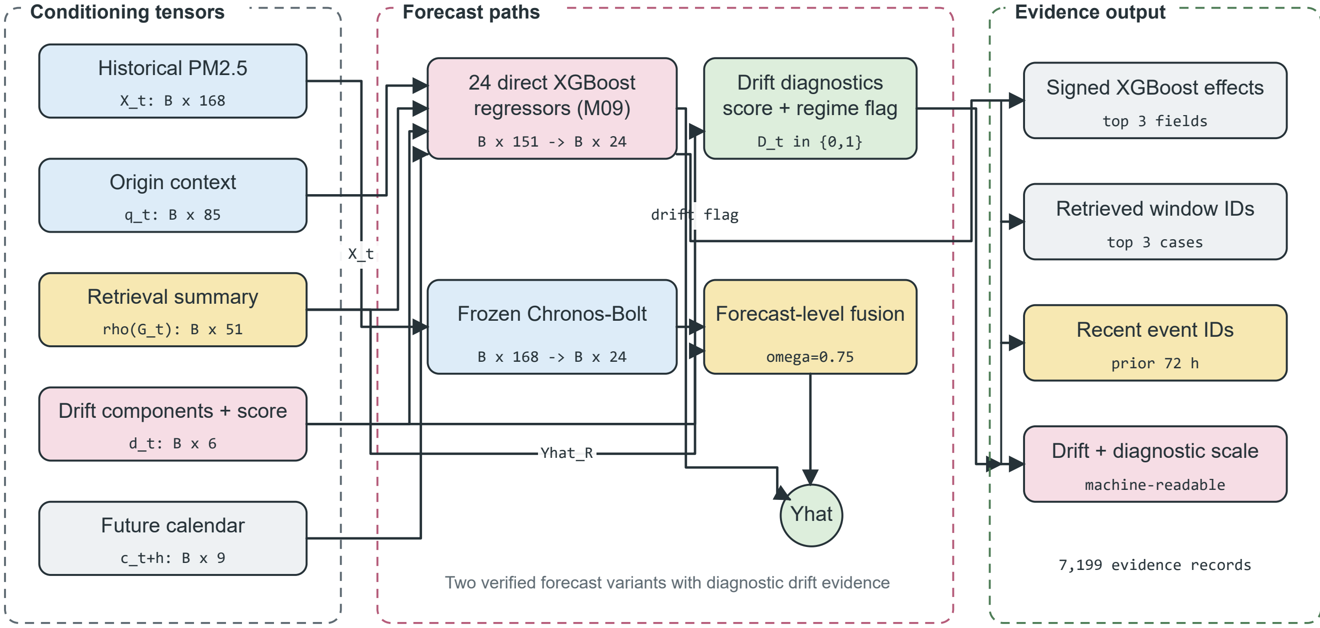


Fig. 4. Drift-Aware Forecast and Evidence Head. Context, retrieval, drift, and future-calendar tensors feed direct horizon regressors; the frozen Chronos-Bolt path is fused only at forecast level. Stored feature effects, retrieved cases, event identifiers, drift components, and uncertainty values form the evidence record.

TABLE II
METHODOLOGICAL COMPARISON OF TIMERAF AND EVENT-TIMERAF

| Aspect | TimeRAF | Event-TimeRAF |
|---|---|---|
| Primary objective | Improve zero-shot forecasting by augmenting a frozen TSFM with external time-series knowledge. | Evaluate source-audited, event-context retrieval for local PM2.5 forecasting under temporal shift. |
| Forecasting domain | Multi-domain benchmark datasets. | Los Angeles County hourly PM2.5, 2019–2024. |
| Knowledge base | Curated, consistently normalized time-series windows. | 1,521 training-history windows at the primary 24-hour stride, with denser and sparser sensitivity settings, PM2.5, weather, calendar, and retrospective event context. |
| Retriever | Learned MLP dual encoder with forecaster-guided training. | Random, cosine, and transparent hybrid similarity; no learned retriever. |
| Knowledge integration | Internal Channel Prompting of candidate embeddings. | Feature-level XGBoost inputs and forecast-level Chronos-Bolt fusion. |
| Backbone | Frozen TSFM, principally TTM-Base. | Direct XGBoost horizon models and frozen `amazon/chronos-bolt-small`. |
| Leakage control | Non-overlapping knowledge windows and cross-dataset training constraints. | Candidate windows may overlap one another, but every candidate target must end before the query lookback; validation and test use training history only. |
| Event handling | Not an explicit module. | Availability-tagged NOAA Storm Events features and source-record evidence, reported retrospectively. |
| Drift and explanation | Retrieved-case inspection. | Five-component drift audit, signed feature effects, retrieved IDs, event IDs, and diagnostic scale. |
| Implementation scope | Learned retriever and Channel Prompting are implemented. | Those mechanisms are not implemented and remain future extensions. |

Weather covariates were taken from NOAA Global Hourly/ISD through the official NOAA Open Data Dissemination public buckets [35]. The selected station was `72297023129`, Long Beach / Daugherty Field / Airport, 10.28 km from the PM2.5 target centroid. The complete weather-feature coverage after allowed past-only short-gap filling was 0.9942, and the raw measured core coverage was 0.9905.

Environmental events were loaded from hash-verified NOAA Storm Events annual detail archives [36]. Because direct access to NCEI was unavailable in the execution environment, the experiment used a locally staged cache containing unchanged NOAA annual archives and a source manifest with official URLs, byte counts, and SHA-256 hashes. The audit retained the delivery mode `verified_official_noaa_cache`. The event audit found 117 event days, 2,192 source-overlap days, and three qualifying event categories. High wind (76 records), flood

TABLE III
DATA-READINESS AUDIT

| Gate | Observed | Requirement | Status |
|---|---|---|---|
| PM2.5 coverage | 0.9997 | $\geq 0.70$ | Pass |
| Weather coverage | 0.9942 | $\geq 0.95$ | Pass |
| Event days | 117 | $\geq 30$ | Pass |
| Source-overlap days | 2,192 | $\geq 180$ | Pass |
| Qualifying categories | 3 | $\geq 2$ | Pass |
| Strict availability | False | — | Caveat |

(61), and other weather (40) meet the configured 20-record category threshold; heavy rain (18) and wildfire (9) are retained but do not qualify. NOAA Storm Events does not expose machine-readable publication timestamps; therefore, event-aware outputs use the recorded event start time as a retrospective availability assumption. They are not strict real-time event forecasts. Table III reports the resulting readiness status. The source audit and window contract are visualized in Figure 2.

Window construction started from 52,608 aligned hourly rows and 52,417 possible forecast origins. It excluded 48 origins at split boundaries, eight for a missing lookback, 144 for a missing target, and 3,885 for missing origin features; none was excluded for future-calendar values. The remaining 48,332 origins comprise 34,020 training, 7,113 validation, and 7,199 test origins.

Table IV consolidates the evaluated scope and the role of each source in the forecasting pipeline.

## B. Data and Code Availability

The raw public records and download pages are available from EPA AirData, NOAA Global Hourly/ISD, and NOAA Storm Events [34]–[36]. The experiment used a locally staged NOAA Storm Events cache because direct NCEI access was unavailable in the execution environment; every cached NOAA file was verified against the generated source manifest. Source code is available at https://github.com/shasabbir/event-time-raf. The implementation uses Python 3.12.13, NumPy, pandas, scikit-learn, XGBoost, PyTorch, and the `chronos-forecasting` package; the repository contains the configuration, notebooks, source modules, and tests needed to reproduce the pipeline when the public source files are accessible.

## C. Forecasting Task and Split

The forecasting task predicts PM2.5 from one to 24 hours ahead using a 168-hour lookback window. The split is chronological: the earliest 70% of valid windows are used for training, the next 15% for validation, and the latest 15% for testing. The final test set contains 7,199 forecast origins and 172,776 forecasted hourly target points. Test origins span 7 February through 30 December 2024 UTC, and their targets end on 31 December 2024 UTC. Random splitting is not used.

## D. Models

The evaluated models are persistence, daily seasonal naive, weekly seasonal naive, hourly-by-month climatology, ridge regression, direct XGBoost models, random, cosine, calendar-only, and event-stratified retrieval, Event-TimeRAF without and with drift features, frozen Chronos-Bolt, and three validation-tuned Chronos fusion variants. The fusion controls replace retrieval with climatology or persistence while preserving the weight-selection procedure. An auxiliary random-retrieval XGBoost model A01 and an event-free model A00 support the ablation analysis. A01 replaces cosine retrieval features with seeded random retrieval features while retaining the context regressor. A00 omits event features and uses event-free hybrid retrieval and a four-component drift diagnostic. The foundation-model evaluation uses the `amazon/chronos-bolt-small` checkpoint. Table V groups the evaluated configurations by their forecasting and retrieval roles.

## E. Implementation and Hyperparameters

The experiment was executed in a cloud-hosted Linux 6.12.90 environment using Python 3.12.13. The runtime was 3,319.95 seconds. Key package versions recorded in the run manifest include NumPy 2.0.2, pandas 2.3.3, scikit-learn 1.6.1, XGBoost 3.2.0, PyTorch 2.10.0+cu128, and chronos-forecasting 2.3.1. Table VI reports the principal hyperparameters.

## F. Evaluation Metrics and Statistical Checks

The primary metrics are MSE, MAE, RMSE, MAPE, sMAPE, and $R^2$. MSE is retained for comparability with the TimeRAF evaluation, while MAE and RMSE provide more interpretable PM2.5 error scales. For $n$ observed forecast values $y_j$ and predictions $\hat{y}_j$, the principal error metrics are

$$\begin{aligned} \mathrm{MSE} &= \frac{1}{n}\sum_{j=1}^{n}(y_j - \hat{y}_j)^2, \\ \mathrm{MAE} &= \frac{1}{n}\sum_{j=1}^{n}|y_j - \hat{y}_j|, \\ \mathrm{RMSE} &= \sqrt{\mathrm{MSE}}, \end{aligned} \tag{20}$$

and explained variance is summarized by

$$R^2 = 1 - \frac{\sum_{j=1}^{n}(y_j - \hat{y}_j)^2}{\sum_{j=1}^{n}(y_j - \bar{y})^2}. \tag{21}$$

Equation (20) is aggregated across all forecast origins and horizons for the overall result, while (21) uses the same set of points. MAPE and sMAPE are retained in machine-readable outputs but are not used for the main ranking because low PM2.5 values can make percentage errors unstable. Subset metrics are reported only when at least 50 forecast origins are available. The event subset contains 532 origins, the active-event subset contains 293 origins, and the drift subset contains 566 origins, so all pre-specified subsets satisfy the minimum sample-size criterion. Primary pairwise comparisons use 2,000 paired block-bootstrap resamples with 168-origin

TABLE IV
DATASET AND SOURCE SUMMARY

| Data group | Source | Evaluated scope | Use in pipeline |
|---|---|---|---|
| PM2.5 | EPA AirData/AQS hourly parameter 88101 | Los Angeles County, 2019–2024; 52,602 observed target hours | Target sequence and lag/rolling features |
| Weather | NOAA Global Hourly/ISD, station `72297023129` | 2019–2024; 0.9942 usable core coverage | Temperature, humidity, wind, pressure, and rain features |
| Events | NOAA Storm Events annual detail archives | 204 Los Angeles records; 117 event days | Retrospective event-start features and explanation evidence |
| Calendar | Generated from timestamps and public holiday rules | 2019–2024 hourly calendar | Hour, weekday, weekend, month, season, and holiday features |

TABLE V
EVALUATED MODEL FAMILIES

| ID | Description |
|---|---|
| M00–M02 | Persistence and seasonal naive baselines |
| M03–M04 | XGBoost PM2.5-only and context models |
| M05–M06 | Random and cosine retrieval-only baselines |
| M07 | XGBoost with cosine retrieval features |
| M08–M09 | Event-TimeRAF without/with drift indicators |
| M10–M11 | Frozen Chronos-Bolt and retrieval fusion |
| C00–C03 | Climatology, ridge, calendar, and event-stratified controls |
| C10–C11 | Chronos climatology and persistence fusion controls |

blocks. Diebold–Mariano tests use a Newey–West/HAC variance estimate with lag 167, followed by Holm adjustment within each pre-specified metric family.

### G. Comparison with Existing Studies

Direct numerical comparison with existing TSFM and air-quality papers is not valid because they use different datasets, forecast horizons, variables, and evaluation protocols. Table VII therefore compares methodological coverage rather than claiming cross-paper performance dominance.

## VIII. RESULTS AND DISCUSSION

### A. Overall Comparison with Baselines

Table VIII reports the test-set results. The strongest pre-specified supervised result is M04, which uses PM2.5 history, weather, and calendar features and obtains MSE 26.185, MAE 3.125, RMSE 5.117, and $R^2$ 0.379. The full feature-level Event-TimeRAF model M09 obtains MSE 26.734 and MAE 3.144. Thus, the complete event, retrieval, and drift feature set does not improve all-origin error over the simpler context model.

The frozen Chronos-Bolt evaluation completed successfully. Validation selected a retrieval-fusion weight of 0.75. On the test set, M11 changes MSE from 28.941 to 28.733. The M11–M10 difference is -0.209 with a 95% block-bootstrap interval $[-0.654, 0.216]$ and Holm-adjusted DM $p = 1.000$, so the change is unresolved. The matched climatology fusion C10 obtains MSE 27.450. M11 is worse than C10 by 1.283 MSE, with interval $[0.694, 1.913]$ and adjusted $p < 0.001$. The persistence control selects a TSFM weight of 1.0 and therefore equals M10. Consequently, this experiment does not isolate a retrieval-specific benefit; ordinary validation-tuned forecast combination performs better.

### B. Ablation Study

Table IX summarizes component-wise comparisons. Weather and calendar context reduces MSE by 3.764 relative to PM2.5-only XGBoost, and cosine retrieval improves substantially over random retrieval when retrieval is used alone. The remaining feature-level additions have intervals that cross zero. In particular, M09 is 0.550 MSE worse than M04, and its comparison with event-free A00 is also unresolved. Drift indicators change M08 by only 0.034 MSE. These results do not support a monotone accuracy contribution from each named module.

The intervals are paired block-bootstrap intervals. Holm-adjusted DM values are shown only for the pre-specified primary family; dashes denote secondary diagnostic comparisons rather than missing experiment runs.

The $k$ sensitivity results in Table X show that the retrieval-only cosine baseline improves as more neighbors are used within the tested range. The best retrieval-only configuration among $k \in \{1, 4, 8, 16\}$ is $k = 16$, with MSE 34.440. The reported Event-TimeRAF feature models used the pre-specified default $k = 8$.

### C. Knowledge-Base Density and Event Sensitivity

Table XI shows that the original sparse-memory concern is consequential for retrieval alone. Reducing the stride from 192 to six hours increases candidate count from 191 to 5,644 and lowers retrieval-only MSE from 39.037 to 34.616. A one-hour stride is not better than six hours. The separately trained full feature model also reaches its lowest observed MSE at six hours (26.412), but it remains behind M04 (26.185). Dense retrieval therefore improves the historical analogue baseline without overturning the main model ranking.

The event channel is active but not beneficial under the tested representation. At weight 0.2 it changes at least one top-eight candidate for 91.4% of test queries and raises the event-candidate fraction for event-context queries from 32.9% to 95.7%. Nevertheless, Table XII shows that all-origin and event-subset errors increase as event weight rises. Event-stratified retrieval is also worse than the event-free ranking. The supported conclusion is therefore that this count-based event representation and ranking rule are ineffective, not that environmental events contain no forecast signal.

TABLE VI
EXPERIMENTAL HYPERPARAMETERS AND RUNTIME SETTINGS

| Component | Setting |
|---|---|
| Study period | 2019–2024 |
| Lookback and horizon | $L = 168$ hours, $H = 24$ hours |
| Split | 70% train, 15% validation, 15% test, chronological |
| SACB feature groups | 23 PM2.5, 14 weather, 9 calendar, 39 event features |
| SACB missing-data rule | Past-only filling for gaps up to 3 hours; no target interpolation |
| LSER retrieval | $k = 8$, sensitivity for $k \in \{1, 4, 8, 16\}$ |
| LSER knowledge base | 1,521 windows at 24-hour primary stride; sensitivity at 192, 24, 6, and 1 hours |
| LSER similarity weights | time series 0.5, weather 0.2, calendar 0.1, event 0.2 |
| LSER event-weight sensitivity | $\delta \in \{0, 0.2, 0.5, 0.8, 1.0\}$ with remaining weights renormalized |
| DFEH direct input | 151 features for each of 24 horizon regressors |
| DFEH XGBoost | 350 estimators, max depth 6, learning rate 0.05, subsample 0.85, colsample 0.85, $\lambda = 1.0$ |
| DFEH drift detector | 5 components; 24-hour recent window, 168-hour reference window, 0.90 validation quantile threshold |
| DFEH Chronos path | `amazon/chronos-bolt-small`, batch size 64 |
| DFEH fusion weights | $\{0, 0.25, 0.5, 0.75, 1.0\}$; selected $\omega = 0.75$ |
| Explanation record | Top 3 feature effects, windows, and recent event IDs |
| Inference | 2,000 paired block-bootstrap resamples, 168-origin blocks; DM HAC lag 167; Holm adjustment |

TABLE VII
METHODOLOGICAL COMPARISON WITH RELATED FORECASTING WORK

| Method family | TSFM | Retrieval | Events | Explainability | Comment |
|---|---|---|---|---|---|
| PatchTST-style Transformers [22] | No | No | No | Limited | Strong sequence modeling, but event context is not central. |
| Chronos [5] | Yes | No | No | Limited | Frozen zero-shot baseline used in this study. |
| TimeRAF [9] | Yes | Yes | No | Retrieval cases | Base retrieval-augmented TSFM method; does not target official event-aware PM2.5. |
| XGBoost context baseline [10] | No | No | No | Feature importance | Strong supervised tabular baseline for local PM2.5. |
| Event-TimeRAF | Yes | Yes | Yes | Evidence-grounded | Adds official event audit, temporal leakage control, drift indicators, and Chronos retrieval fusion. |

TABLE VIII
MANIFEST-BACKED OVERALL TEST RESULTS FOR 24-HOUR PM2.5 FORECASTING

| Model | MSE | MAE | RMSE | $R^2$ |
|---|---|---|---|---|
| C00 hour–month climatology | 36.297 | 4.038 | 6.025 | 0.139 |
| C01 ridge context | 28.448 | 3.226 | 5.334 | 0.325 |
| C02 calendar retrieval | 45.534 | 4.429 | 6.748 | -0.080 |
| C03 event-stratified retrieval | 38.428 | 3.924 | 6.199 | 0.089 |
| M00 persistence | 47.649 | 4.137 | 6.903 | -0.130 |
| M01 daily seasonal | 48.706 | 4.057 | 6.979 | -0.155 |
| M02 weekly seasonal | 65.606 | 5.123 | 8.100 | -0.556 |
| M03 XGBoost PM2.5 | 29.949 | 3.368 | 5.473 | 0.290 |
| M04 XGBoost context | **26.185** | **3.125** | **5.117** | **0.379** |
| M05 random retrieval | 44.089 | 4.346 | 6.640 | -0.046 |
| M06 cosine retrieval | 36.243 | 3.818 | 6.020 | 0.140 |
| M07 XGBoost cosine | 26.442 | 3.132 | 5.142 | 0.373 |
| M08 Event-TimeRAF no drift | 26.700 | 3.144 | 5.167 | 0.367 |
| M09 Event-TimeRAF full | 26.734 | 3.144 | 5.171 | 0.366 |
| M10 frozen Chronos-Bolt | 28.941 | 3.205 | 5.380 | 0.314 |
| M11 Chronos + retrieval | 28.733 | 3.205 | 5.360 | 0.319 |
| C10 Chronos + climatology | 27.450 | 3.137 | 5.239 | 0.349 |
| C11 Chronos + persistence | 28.941 | 3.205 | 5.380 | 0.314 |

### D. Event and Drift Subsets

Table XIII reports representative event and drift subsets for the strongest context model, the full Event-TimeRAF model, and the two Chronos variants. All subsets satisfy the pre-specified minimum sample-size criterion. Event targets have variance 22.636, compared with 43.346 for non-event targets, while drift and non-drift target variances are 69.593 and 39.826. Raw subset MSE therefore reflects both model error and target difficulty. The drift subset is defined by the validation-calibrated DFEH flag and is used only for stratified evaluation.

### E. Horizon-Wise and Retrieval Diagnostics

Fig. 5 and Fig. 6 show horizon-wise test errors. Fig. 7 summarizes retrieval behavior, and Fig. 8 shows the drift-score distribution across the test period. Figure 9 shows a representative

TABLE IX
SELECTED PAIRED BLOCK-BOOTSTRAP ABLATION RESULTS

| Comparison | MSE diff. | 95% CI low | 95% CI high | Holm DM $p$ |
|---|---|---|---|---|
| M04 minus M03, weather/calendar | -3.764 | -7.120 | -1.709 | 0.067 |
| M06 minus M05, cosine/random retrieval | -7.846 | -11.333 | -4.849 | — |
| M07 minus M04, retrieval summary | 0.258 | -0.269 | 1.012 | — |
| M08 minus M07, event-hybrid ranking | 0.258 | -0.087 | 0.690 | — |
| M09 minus M08, drift indicators | 0.034 | -0.079 | 0.147 | — |
| M09 minus A00, event context | 0.216 | -0.114 | 0.635 | 1.000 |
| M09 minus M04, complete feature path | 0.550 | -0.122 | 1.592 | 1.000 |
| M11 minus M10, retrieval fusion | -0.209 | -0.654 | 0.216 | 1.000 |
| M11 minus C10, climatology placebo | 1.283 | 0.694 | 1.913 | $< 0.001$ |

TABLE X
COSINE RETRIEVAL SENSITIVITY TO $k$

| $k$ | MSE | MAE | RMSE |
|---|---|---|---|
| 1 | 60.555 | 4.886 | 7.782 |
| 4 | 40.004 | 4.010 | 6.325 |
| 8 | 36.243 | 3.818 | 6.020 |
| 16 | 34.440 | 3.711 | 5.869 |

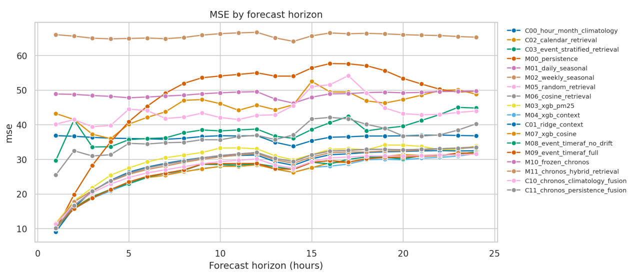


Fig. 5. MSE by forecast horizon on the Los Angeles County test set.

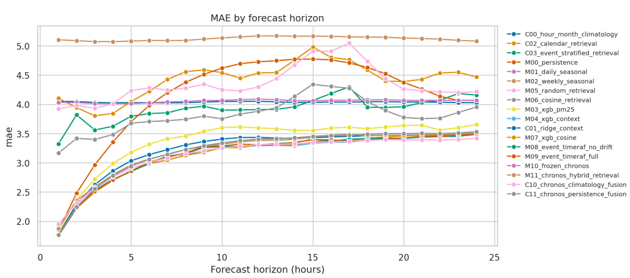


Fig. 6. MAE by forecast horizon on the Los Angeles County test set.

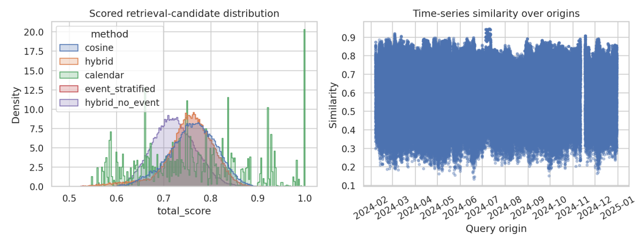


Fig. 7. Similarity and candidate diagnostics for test-set retrieval.

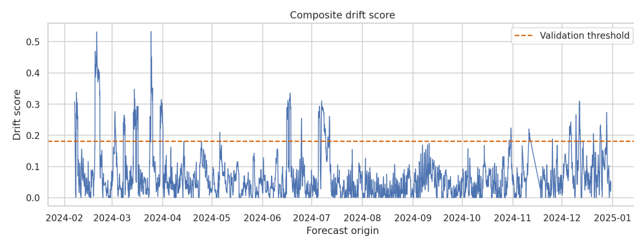


Fig. 8. Composite drift-score diagnostics on the test set.

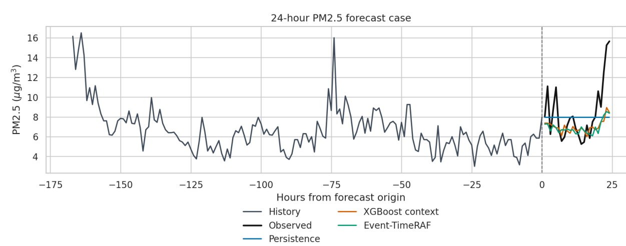


Fig. 9. Representative 24-hour forecast case from the test set.

test-origin forecast, allowing M04 and M09 to be compared with the observed target. These plots are regenerated from predictions identified by `20260810T103436161252Z`. The validation-calibrated upper-tail drift threshold flags 712 of 7,113 validation origins (10.0%) and 566 of 7,199 test origins (7.9%). This difference is reported as a diagnostic property; the flag does not switch models or establish that concept drift occurred.

## *F. Evidence and Traceability Analysis*

Within the DFEH, traceability is implemented as a source-grounded evidence record rather than as a post-hoc free-text interpretation. For each evaluated forecast origin, the pipeline stores the run identifier, window identifier, origin timestamp, retrieved historical evidence identifiers, event evidence identifiers, leading feature effects, drift components, retrieval spread, validation residual RMSE, diagnostic uncertainty scale, drift score, drift flag, and event-availability mode. The evaluation produced 7,199 explanation records in `explanations.parquet`. A representative record identifies the strongest local contribution fields as the current PM2.5 value and recent rolling PM2.5 means, links the prediction to three retrieved historical windows, links recent source-recorded storm events when available, and marks distribution shift when the drift detector is active.

Table XIV summarizes the explanation layer. This design

TABLE XI
KNOWLEDGE-BASE STRIDE SENSITIVITY ON THE TEST SET

| Stride (h) | Candidates | Retrieval MSE | Retrieval MAE | Full M09 MSE |
|---|---|---|---|---|
| 192 | 191 | 39.037 | 3.991 | 26.712 |
| 24 | 1,521 | 38.411 | 3.921 | 26.734 |
| 6 | 5,644 | **34.616** | **3.660** | **26.412** |
| 1 | 34,020 | 36.416 | 3.790 | 26.740 |

TABLE XII
EVENT-CHANNEL SENSITIVITY FOR RETRIEVAL-ONLY FORECASTS

| Method | Event weight | Top-$k$ changed | All-origin MSE | Event-subset MSE |
|---|---|---|---|---|
| Hybrid | 0.0 | 0.000 | **38.080** | **14.299** |
| Hybrid | 0.2 | 0.914 | 38.411 | 16.401 |
| Hybrid | 0.5 | 0.914 | 38.703 | 19.292 |
| Hybrid | 0.8 | 0.914 | 38.969 | 21.201 |
| Hybrid | 1.0 | 1.000 | 40.860 | 27.130 |
| Event-stratified | 0.2 | 0.917 | 38.428 | 16.638 |

TABLE XIII
REPRESENTATIVE EVENT AND DRIFT SUBSET RESULTS

| Subset | Model | Origins | MSE | MAE |
|---|---|---|---|---|
| Non-event | M04 XGBoost context | 6,667 | 27.520 | 3.179 |
| Event | M04 XGBoost context | 532 | 9.451 | 2.450 |
| Non-event | M09 Event-TimeRAF full | 6,667 | 27.995 | 3.187 |
| Event | M09 Event-TimeRAF full | 532 | 10.932 | 2.613 |
| Non-drift | M04 XGBoost context | 6,633 | 26.060 | 3.086 |
| Drift | M04 XGBoost context | 566 | 27.650 | 3.585 |
| Non-drift | M09 Event-TimeRAF full | 6,633 | 26.282 | 3.085 |
| Drift | M09 Event-TimeRAF full | 566 | 32.039 | 3.833 |
| Drift | M10 frozen Chronos-Bolt | 566 | 29.539 | 3.524 |
| Drift | M11 Chronos + retrieval | 566 | 29.210 | 3.610 |

TABLE XIV
EXPLANATION EVIDENCE STORED BY EVENT-TIMERAF

| Evidence field | Interpretation role |
|---|---|
| Top feature effects | Local model drivers from the direct forecaster. |
| Retrieved windows | Historical analogues used by the retrieval module. |
| Event evidence IDs | Official storm-event context linked to the origin. |
| Drift components | Mean, variance, weather, similarity, and event-burst shift signals. |
| Diagnostic scale | Combined retrieval spread and validation residual RMSE; not a calibrated interval. |

TABLE XV
COMPUTATIONAL AND REPRODUCIBILITY SUMMARY

| Item | Recorded value |
|---|---|
| Run identifier | `20260810T103436161252Z` |
| Runtime | 3,319.95 seconds |
| Platform | Cloud-hosted Linux environment |
| Python | 3.12.13 |
| Forecast horizon | 24 hours |
| Lookback window | 168 hours |
| Bootstrap resamples | 2,000; block length 168 origins |
| Chronos checkpoint | `amazon/chronos-bolt-small` |
| Code availability | https://github.com/shasabbir/event-time-raf |

supports auditability because every explanatory statement can be traced back to a model feature, a retrieved time-series window, an official event record, or a drift component. It does not prove causality: event evidence is reported as contextual support, and the drift flag is treated as an uncertainty warning.

Chronos-Bolt's nominal 80% interval covers 78.2% of test targets, with mean width 9.890 $\mu$g/m$^3$. This interval result is distinct from the `diagnostic_uncertainty_scale` in the evidence record, which combines retrieval spread and validation residual RMSE and is not a calibrated interval.

### G. Computational Cost and Reproducibility

The complete experiment executed in 3,319.95 seconds in a cloud-hosted Linux environment using Python 3.12.13. The recorded package versions include NumPy 2.0.2, pandas 2.3.3, scikit-learn 1.6.1, XGBoost 3.2.0, PyTorch 2.10.0+cu128, and `chronos-forecasting` 2.3.1. The run manifest records the configuration SHA-256 hash and artifact checksums. The CPU, GPU, and RAM model identifiers were not saved in the manifest; therefore this paper reports runtime and software versions but does not claim hardware-normalized efficiency. Table XV lists the reproducibility fields retained by the archived run.

The main interpretation is that conventional supervised context features are strong for this Los Angeles County task. Dense historical memory improves the retrieval-only baseline, but retrieval, event, and drift features do not beat M04. The small M11 change relative to frozen Chronos is unresolved, and the climatology fusion control is significantly better than M11. Event-TimeRAF is therefore evidence for an auditable evaluation protocol and a scoped negative result, not for retrieval-specific or event-specific superiority.

Finally, all event-aware interpretations must be read with the availability caveat. NOAA Storm Events records are official and hash-verified, but the source does not provide machine-readable publication timestamps. The experiment records event start as the availability time, so event-related results are retrospective sensitivity results. A strict operational study would require event or smoke sources with genuine issue or publication times, such as an audited NOAA HMS cache or alert product archive.

## IX. Limitations

The first limitation is event availability. NOAA Storm Events records are official and hash-verified, but the public detail files do not provide machine-readable publication timestamps. The SACB therefore uses event start as an availability proxy, making event-conditioned results retrospective sensitivity analyses rather than operational forecasts.

The second limitation is geographic validation. The experiment covers one county-level PM2.5 aggregate from 2019 through 2024. It does not establish transfer to another county, monitoring topology, climate, or emissions regime, and therefore does not satisfy the external-validation standard for a mature generalization claim.

The third limitation concerns model integration. The LSER uses transparent similarity functions rather than a learned retriever, and the DFEH combines Chronos-Bolt and retrieval only at forecast level. Event-TimeRAF therefore does not reproduce TimeRAF's learned dual encoder or internal Channel Prompting.

The fourth limitation concerns computational reporting. Hardware model identifiers and peak memory were absent from the run manifest, so runtime cannot be interpreted as a hardware-normalized efficiency result.

The fifth limitation is target aggregation. The hourly county median combines a changing set of monitors and suppresses localized extremes. This improves series coverage but can weaken event associations and prevents conclusions about individual neighborhoods or monitoring sites.

## X. Future Work

Operational event studies should replace event-start proxies with archives that expose genuine issue or publication times and should evaluate source types more closely tied to PM2.5, such as smoke observations. External validation should apply the unchanged protocol to a second US county with distinct meteorology and emissions. Longer-term work may evaluate site-level targets, a learned event-context retriever, richer event embeddings, calibrated probabilistic uncertainty, and internal TSFM conditioning. Each extension should retain the source audit, temporal embargo, and component-wise controls used here.

## XI. Conclusion

This study audited event-aware retrieval for 24-hour Los Angeles County PM2.5 forecasting. The pipeline aligns official environmental records, enforces a query–candidate temporal embargo, evaluates memory density and event-channel sensitivity, and retains traceable evidence. M04 is the strongest evaluated model, with MSE 26.185, MAE 3.125, RMSE 5.117, and $R^2$ 0.379; full M09 obtains MSE 26.734. M11 changes frozen Chronos-Bolt MSE from 28.941 to 28.733, but the paired interval crosses zero, while climatology fusion reaches 27.450 and significantly outperforms M11. Dense retrieval improves the retrieval-only path, whereas the tested event channel does not improve it. The contribution is thus a reproducible evaluation and a bounded negative result: historical retrieval can be implemented under a verified query–candidate temporal embargo and audited thoroughly without necessarily adding forecast value beyond strong local context and simple forecast combination.

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